

AI & ML Task 3 Result Report

Objective

Validate regression models, control overfitting, tune hyperparameters and select the best model.

Baseline Results

Linear Regression: RMSE 0.7456, R^2 0.5758, CV R^2 0.5530

Ridge Regression: RMSE 0.7456, R^2 0.5758, CV R^2 0.5530

Decision Tree: RMSE 0.7037, R^2 0.6221, CV R^2 0.3556

Hyperparameter Tuning

Best Parameters:

max_depth = 10

min_samples_leaf = 4

min_samples_split = 2

Best CV Score = 0.7101

RMSE = 0.6391

Test R^2 = 0.6883

Training R^2 = 0.8229

Conclusion

The tuned Decision Tree Regressor achieved the lowest RMSE, highest R^2 score and strongest cross-validation performance. Hyperparameter tuning reduced overfitting and improved generalization. Therefore, the tuned Decision Tree is selected as the final model.